10TH CLASS MATHEMATICS — NEW SYLLABUS (PECTAA)

Exercise 10.1 — Construction of Circles and their Centres

Unit 10 | Solved Exercise | New Book (PECTAA)

Exercise 10.1 (Solved)

These are construction questions, so the full-mark answer is the written construction: Given, Required, then the numbered steps. Draw the figure with a sharp pencil, a pair of compasses and a ruler while you write each step.

1. Construct a circle with the help of given radius and verify its centre by construction. (i) r = 1.5 cm (ii) r = 1.7 cm (iii) r = 2 cm

Construction for r = 1.5 cm (the same steps for 1.7 cm and 2 cm)

1. Draw a line segment OP of length 1.5 cm.
2. Take O as centre and open the compasses 1.5 cm.
3. Draw the complete circle. This circle has centre O and radius 1.5 cm.
4. Verification: draw any two chords AB and CD of the circle.
5. Draw the right bisector of AB and the right bisector of CD.
6. Both bisectors meet at the same point, and that point is O, the centre of the circle.

Answer: Circle constructed; its centre is verified as O

2. Take any three non-collinear points P, Q, R and construct a circle passing through these points.

1. Plot the three points P, Q and R so that they do not lie on one straight line.
2. Join P to Q and Q to R.
3. Draw the right bisector of PQ.
4. Draw the right bisector of QR.
5. The two right bisectors cut each other at a point O.
6. Take O as centre and radius mOP = mOQ = mOR, and draw the circle.

Answer: The circle through P, Q and R is drawn with centre O

3. Draw an arc ABC and complete a circle by finding its centre.

1. Draw the given arc ABC.
2. Join A to B and B to C, these are two chords of the circle.
3. Draw the right bisector of AB.
4. Draw the right bisector of BC.
5. Let the two right bisectors meet at O; O is the centre of the circle.
6. Take O as centre and radius OA = OB = OC, and draw the complete circle.

Answer: The arc is completed into a circle with centre O

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4. Draw an arc PQR and complete a circle without finding its centre.

1. Take AB and BC as two equal chords of the given arc ABC, so mAB = mBC.
2. Produce AB towards P and BC towards Q to form the rays AP and BQ.
3. At A construct an angle b equal to the angle a at C, that is m∠b = m∠a.
4. Take mCD = mAB = mBC and construct m∠c = m∠a = m∠b.
5. In the same way construct ∠d and ∠e on the other side.
6. Join the points so obtained; the curve so drawn is the rest of the circle and the arc ABC now becomes a part of it.

Answer: The circle is completed from the arc ABC without locating the centre

5. Take any three non-collinear points (locations of the lamp posts in the park), construct a circle passing through these points.

1. Mark the three lamp posts as the points A, B and C (not in a straight line).
2. Join AB and BC.
3. Draw the right bisector of AB.
4. Draw the right bisector of BC.
5. The two bisectors meet at O, which is equidistant from all three posts.
6. Draw the circle with centre O and radius OA = OB = OC.

Answer: The circular path through the three lamp posts is drawn

6. A part of the Ferris wheel is visible as an arc. Using any three points on the arc, construct the circle by finding its centre.

1. Take three points A, B and C on the visible arc.
2. Join AB and BC.
3. Draw the right bisectors of AB and of BC.
4. The bisectors meet at the centre O of the wheel.
5. Draw the full circle with centre O and radius OA = OB = OC.

Answer: The centre of the Ferris wheel is located and the circle completed

7. ABC is an arc of a fountain, complete a circle without finding its centre.

1. Take two chords AB and BC of the arc ABC with mAB = mBC.
2. Produce AB and BC to form the rays.
3. Construct m∠b = m∠a at A, as shown in the figure.
4. Take mCD = mAB = mBC and construct m∠c = m∠a = m∠b.
5. Repeat the same angles and equal chords round the arc.
6. Join the points obtained; the complete circular outline of the fountain is ready.

Answer: The circle is completed from the arc ABC without finding the centre

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